NIC project proposal

1. Project Title

PheroSAM-3D: A Hybrid Multi-Colony Ant Framework for Consistent and Correctable Multi-Organ Volumetric Segmentation with Medical Foundation Models

In one sentence: colonies of artificial ants, one per organ, search for and refine the box prompts on every slice of a 3D scan, compete so that organs do not overlap, and let a doctor's single correction spread through the whole volume, all on top of a frozen foundation model (MedSAM or MedSAM2) with no extra training.

Item	Choice
Field	Nature-Inspired Computation (NIC) for medical image segmentation
Main NIC algorithm	Hybrid discrete–continuous ACO (MAX–MIN Ant System + ACO_R), seeded by dynamic programming, with competing colonies per organ
Main AI model	MedSAM (frozen) as the main model, MedSAM2 in plug-in mode, and a 2D nnU-Net as the box proposer
Data type	3D CT and MRI volumes from several hospitals
Output	3D multi-organ masks + a short list of slices for a doctor to check, with corrections propagated automatically

Our contributions (what is new in this paper)

1. **Hybrid search, not just selection.** Ants choose a candidate box per slice and then refine its position and size continuously (ACO_R), so results can be better than any box the proposer produced.
2. **DP-seeded ACO with a clear answer to "why not dynamic programming?".** Dynamic programming solves the local part of the problem exactly and seeds the pheromone; the ants then optimize the global terms that dynamic programming cannot handle. We show when and why the ants are needed.
3. **Competing colonies for multi-organ consistency.** One colony per organ; colonies penalize each other for overlapping masks and follow simple anatomical rules (for example, the liver lies above the right kidney). No single-path method can do this jointly.
4. **Correction that spreads.** A doctor fixes one flagged slice; that fix becomes a fixed node in the graph, and the colonies re-optimize the whole volume. We measure Dice versus number of corrections.
5. **Model-agnostic layer.** The same colony can choose which keyframes MedSAM2 should be prompted on, so it improves the strongest baseline instead of only competing with it.

2. Motivation

Doctors need accurate 3D outlines of organs and tumors, but drawing them by hand takes a long time and medical AI models often fail when moved to a new hospital.

Clinical motivation

- Radiotherapy planning, surgery planning and tumor follow-up all need a 3D mask of the organ or lesion. A single CT or MRI scan can have 30 to 300 slices, so manual drawing can take tens of minutes per patient.
- Foundation models such as MedSAM segment almost any structure well, but they need a box on every slice, and they work on one 2D slice at a time.
- Automatic shortcuts (copying the mask from one slice to the next) let one mistake spread through the whole scan. Doctors then have to check and fix many slices.
- Hospitals use different scanners, so a model trained in one hospital often performs worse in another.

Scientific motivation

- Choosing one box per slice so that the whole 3D result is consistent is a path-finding problem on a graph. This is exactly the kind of problem ant colonies solve in nature.
- Most NIC work in medical segmentation still optimizes old-style thresholds or tunes a network before training. Using NIC at deployment time, on top of a frozen foundation model, is much less explored.
- The method needs no labels and no backpropagation at test time, so it can run on a laptop or hospital workstation.

Why this is useful even if it is not perfect: the pheromone map shows which slices the colony disagreed on. The doctor can review only those slices instead of the whole scan.

3. Problem Definition

We want fully automatic 3D segmentation with MedSAM, where each slice gets a good box, neighboring slices agree with each other, and nothing is retrained.

Background in plain words

Term	Simple meaning
Segmentation	Coloring every pixel that belongs to the organ or tumor. The result is a mask.
Volume / slice	A CT or MRI scan is a stack of 2D images (slices), like slices of bread.
MedSAM	A large model that outlines anything in a medical image if you draw a box around it. Works on one slice at a time.
Prompt	The hint given to MedSAM, here a bounding box.
Domain / domain shift	Each hospital or scanner is a domain. Images from a new domain look slightly different and models get worse.
Error accumulation	When each slice's box is copied from the previous slice's mask, one wrong slice ruins all slices after it.
Label-free	We never use the correct answer at test time; we judge quality from indirect signals.

Formal problem

A volume has slices 1 to D. For each slice d, a proposer gives a small set of candidate boxes. We must pick exactly one box per slice (or "no object" on slices without the target). MedSAM turns each picked box into a 2D mask, and stacking the masks gives the 3D result.

The goal is to choose the sequence of boxes that maximizes a label-free quality score of the whole 3D result:

$$\pi^{*} = \arg arg \quad F\left(M\left(b_{1}\right), \ldots, M\left(b_{D}\right)\right)$$

Here π is one choice of box per slice, M(b) is MedSAM's mask for box b, and F rewards confident, stable, smooth and anatomically plausible 3D shapes. Some parts of F only depend on neighbouring slices, but others (total volume, 3D shape) depend on all slices at once. That global part is why simple slice-by-slice methods are not enough.

Two contributions make the problem harder and more realistic.

- **Continuous refinement:** each chosen box b can be shifted and resized by a small offset δ = (dx, dy, dw, dh), so the solution is a candidate index plus a continuous correction per slice.
- **Several organs at once:** for organs o = 1 to O we choose one path π_o per organ, and the score also depends on how the organs relate to each other:

$$\left(\pi_{1}^{*}, \ldots, \pi_{O}^{*}\right) = \arg arg \quad \sum_{o=1}^{O} F_{o}\left(\pi_{o}\right) \; - \; \mu \sum_{o \neq o'} Overlap\left(\pi_{o}, \pi_{o'}\right) \; - \; \nu \, AnatViol\left(\pi_{1}, \ldots, \pi_{O}\right)$$

Overlap counts voxels claimed by two organs; AnatViol counts broken simple rules (for example, spleen center to the left of the liver center). These joint terms connect all organs and all slices, so they cannot be split into independent slice-to-slice pieces.

- **Fixed nodes from a doctor:** when a doctor corrects slice d for organ o, that box becomes fixed and the search must respect it.

Research questions

1. Can a hybrid ant colony (discrete choice + continuous refinement) produce better and more consistent 3D masks than greedy propagation, interpolation, dynamic programming, GA, PSO, simulated annealing and beam search under the same budget, and can it exceed the best box the proposer offered?
2. When do global terms make dynamic programming insufficient, and how much does DP-seeding help the colony?
3. Do competing colonies reduce organ overlap and anatomical errors compared with segmenting each organ separately?
4. How many doctor corrections are needed to reach a target Dice when each correction is propagated by the colony, compared with standard propagation?
5. Does a label-free score correlate well enough with true Dice to guide the search, and is the pheromone-based uncertainty a reliable pointer to wrong slices?
6. Can the colony improve MedSAM2 by choosing its keyframes?

4. Dataset Description

We need public 3D datasets that come from several hospitals or scanners, so we can test both 3D consistency and the pheromone memory across domains. Sizes below are approximate and must be checked on the official pages before use.

Dataset	Modality	Target	Domains	Approx. size	Role
Multi-site Prostate MRI (RUNMC, BMC, I2CVB, UCL, BIDMC, HK)	T2 MRI	Prostate	6 sites	~116 volumes	Main dataset: small, 3D, real scanner shift
M&Ms (Multi-Centre, Multi-Vendor, Multi-Disease Cardiac)	Cine MRI	Left/right ventricle, myocardium	4 vendors, 6 centres	~375 subjects	Second dataset: vendor shift, small structures
AMOS 2022	CT and MRI	15 abdominal organs	Several centres + 2 modalities	~500 CT + ~100 MRI	many organs at once → tests competing colonies and anatomical rules; also CT→MRI shift

To show honestly where the method works and where it struggles, we add one tumor dataset: the Liver task of the Medical Segmentation Decathlon (contrast CT, liver plus liver tumors, about 131 training volumes; size to be checked). Organs have regular shapes that the shape prior can judge; tumors are irregular, so we expect smaller gains there and will report them separately.

Simulated doctor corrections. For the human-in-the-loop experiment, a "correction" is simulated by replacing the box on a flagged slice with the box taken from the true mask, which is the standard way interactive segmentation papers simulate users.

How we will use them

- **Proposer training:** the 2D nnU-Net is trained on one source domain only (e.g. RUNMC for prostate). All other domains are unseen test hospitals.
- **Splits:** fixed, published train/validation/test splits per site. The validation split is used once to set fitness weights and ACO parameters, then they are frozen.
- **Domain order for the memory experiment:** test hospitals arrive in a recurring stream, e.g. B → C → D → B → C → D, to check whether the colony gets faster the second time.
- **Preprocessing:** resample to a common in-plane spacing, intensity windowing for CT and percentile normalization for MRI, resize slices to MedSAM's 1024×1024 input.

Important check — data leakage: MedSAM was trained on over one million public medical image–mask pairs, which may include some of these datasets (AMOS is reported among them by the RadSAM paper).

Before final experiments we must check MedSAM's training list and either exclude overlapping cases or clearly report the overlap. The multi-site prostate data should be checked first because it is the main dataset.

5. Literature Review

5.1 Promptable foundation models

- SAM introduced promptable segmentation with a heavy image encoder and a light mask decoder.
- MedSAM fine-tuned SAM on more than one million medical image–mask pairs and is driven by bounding boxes. It works on 2D slices.
- Medical SAM Adapter showed MedSAM loses about 11 Dice points on BTCV when box accuracy drops, so box quality matters a lot.

5.2 From 2D slices to 3D volumes

- MedSAM2 puts a box on the middle slice and propagates masks in both directions using memory.
- SAMI3D-DW notes that propagation is still slice-based and accumulated errors still need correction across many slices.
- RadSAM reports that error accumulation in volume-level prompting can cost up to 61 points of 3D Dice.
- SAM-Med3D states that slice-by-slice aggregation ignores inter-slice correlation.
- MemSAM-2.5D fixes inconsistency for liver tumors with cross-slice state propagation, but requires training.
- A zero-shot SAM2 CT study tunes inference choices without training and finds performance is still limited by the model's design.
- A 2026 radiotherapy study fills missing slice boxes by linear interpolation between a few manual boxes.
- Classic graph-based optimal surface segmentation solved 3D consistency as a global graph problem long before foundation models.

5.3 Automatic prompting for SAM / MedSAM

- Learned or trained auto-prompting: AutoSAM, MaskSAM, MedSAM-U (uncertainty-guided multi-prompt), ESP-MedSAM, box-predictor add-ons (listed here and here).
- Training-free prompt search: EviPrompt and multiple-point MedSAM prompting, both working per 2D image.
- RL and GA prompt optimization exist for 2D images: TEPO (reinforcement learning).

5.4 NIC in medical segmentation

- Most ACO work does multi-level thresholding, e.g. enhanced ACO for COVID-19 X-rays. This direction is saturated.
- EvoU-Net used a GA to design a U-Net for prostate MRI; PSO-based transfer learning tunes U-Net training. Both act before deployment.

5.5 Memory across domains (test-time adaptation)

- FOA uses CMA-ES for forward-only test-time adaptation of ViT classifiers.
- DPCore keeps a coreset of prompts per domain and reuses or creates prompts as domains recur. This is why we keep memory inside the ant colony instead of building a separate immune module.

5.6 Background for the additions

- **ACO for continuous problems:** ACO_R (Socha and Dorigo, European Journal of Operational Research, 2008) keeps an archive of good solutions and samples new ones from Gaussian kernels around them. We use it to refine box coordinates.
- **Multi-colony ACO:** several colonies that share or compete for pheromone have been used for multi-objective and partitioning problems (e.g. Middendorf et al., Journal of Heuristics, 2002). We did not find competing colonies used for multi-organ segmentation.
- **Combining exact methods with metaheuristics ("matheuristics"):** seeding a metaheuristic with an exact solution of a simpler sub-problem is an established idea in operations research; we apply it so dynamic programming and ACO cooperate instead of compete.
- **Interactive 3D segmentation:** SAMI3D-DW and RadSAM (Section 5.2) evaluate quality over successive user corrections. They propagate corrections through trained models; ours propagates them through a training-free search.
- **Multi-organ consistency:** most multi-organ methods get consistency from a single trained network that predicts all labels together; training-free methods that segment organs one by one can produce overlapping masks.

5.7 Research gap

Approach	Training-free	Global 3D consistency	Can beat proposer boxes	Multi-organ joint consistency	One correction fixes many slices	Points to slices to review
MedSAM per slice with proposer boxes	Yes	No	No	No	No	No
MedSAM2 / SAM2 propagation	Yes	No (greedy, drifts)	No	No	Partly	No
Box interpolation	Yes	Partly	No	No	Partly	No
Dynamic programming over slices	Yes	Local terms only	No	No	Yes	No
MemSAM-2.5D, SAM-Med3D	No	Yes	Yes	Needs retraining	Partly	Partly
ACO thresholding	Yes	No	No	No	No	No
PheroSAM-3D v2 (ours)	**Yes**	**Yes**	**Yes (ACO_R refinement)**	**Yes (competing colonies)**	**Yes**	**Yes**

6. Selected NIC Algorithms

The main algorithm is a hybrid MAX–MIN Ant System: discrete box choice plus continuous refinement (ACO_R), seeded by dynamic programming, with one competing colony per organ. GA, PSO, simulated annealing, beam search, dynamic programming and simple heuristics are the comparison methods.

6.1 How a real ant colony works (the inspiration)

Ants searching for food leave a chemical called pheromone on their path. Short, good paths are walked more often, so they collect more pheromone and attract more ants. Pheromone also evaporates over time, so bad paths fade. Without any leader, the colony settles on the best path. Communicating through marks left in the environment is called stigmergy.

6.2 Main algorithm: MAX–MIN Ant System

- Each ant walks from the first slice to the last and picks one candidate box per slice. Its full walk is one complete 3D segmentation.
- Its choice at each slice depends on two things: pheromone (what earlier ants learned) and heuristic information (how good the box looks on its own).
- After each round, only the best ant adds pheromone, and all pheromone evaporates a little.
- Pheromone is kept between a minimum and a maximum value. This stops the colony from locking onto a wrong path too early, which matters in noisy medical images.

6.3 Extension 1: hybrid discrete–continuous ants (fixes the proposer ceiling)

An ant could only pick one of the proposer's boxes, so the result could never be better than the best candidate. Now each ant does two things per slice:

1. **Pick** a candidate box with the MMAS rule (discrete part).
2. **Refine** it with a small offset δ = (dx, dy, dw, dh) sampled ACO_R-style: the colony keeps an archive of the best offsets found so far for that slice and draws new ones from Gaussian kernels around them.

Refined boxes are decoded by MedSAM's light decoder only when they are new, and results are cached. Simply put: the ants first choose the right neighbourhood, then nudge the box until the mask looks best.

6.4 Extension 2: DP-seeded ACO (answers "why not dynamic programming?")

- Dynamic programming (Viterbi) finds the exact best path for the **local** part of the score (confidence plus slice-to-slice smoothness) in a fraction of a second.
- That path is used to **seed** the pheromone: its edges start near τ_{max}, the rest at a lower level.
- The ants then improve the **global** part (3D shape, total volume, organ overlap, anatomical rules) that dynamic programming cannot represent exactly.
- Planned analysis: gradually increase the weight of the global terms and show that the gap between DP alone and DP-seeded ACO grows. This turns the strongest baseline into evidence for the method.

6.5 Extension 3: competing colonies for several organs

In nature, neighbouring ant colonies defend their own territory. We copy this:

- One colony per organ, each with its own pheromone and its own slice graph.
- Colonies run in alternation. When an ant of colony o scores its path, the masks currently held by the best ants of the other colonies are fixed; overlapping voxels are penalised.
- **Repulsive pheromone:** a box region strongly claimed by one colony lowers the attractiveness of the same region for the others.
- **Anatomical rules** are simple, checkable relations (left/right, above/below, contains/does not contain) written once per dataset. Breaking a rule is penalised in the joint score.
- Organs with two parts on one slice (for example, the two kidneys) get two colonies, one per side, which also removes the "one box per slice" limitation.

6.6 Minor component: pheromone memory

After a patient is finished, the final pheromone pattern is stored together with a short "fingerprint" of the hospital's images. When a new patient arrives, the closest stored pattern is used to initialize the colony. Same hospital → faster and more reliable search; new hospital → start fresh and store a new pattern.

6.7 Why ACO and not another algorithm

Property needed	ACO (ours)	GA	PSO	Dynamic programming
Builds a solution step by step along a path	Natural fit	Indirect	Indirect	Yes
Uses prior knowledge (MedSAM confidence) while building	Yes, heuristic term	No	No	Yes
Handles whole-volume (non-local) scores	Yes	Yes	Yes	Only approximately
Mixes discrete choice with continuous refinement	Yes (MMAS + ACO_R)	Possible, awkward	Continuous only	No
Several organs searched jointly with competition	Yes (competing colonies)	One big chromosome	One big swarm	No
Accepts fixed nodes from a doctor and re-optimizes	Yes, by clamping pheromone	Yes, by freezing genes	Yes	Yes, local terms only
Leaves an interpretable map of agreement	Yes, pheromone	No	No	No

6.8 Comparison methods (same number of MedSAM decoder calls)

- **Greedy propagation:** each slice uses the box derived from the previous slice's mask.
- **Best box per slice:** the most confident box on every slice, chosen independently.
- **Dynamic programming (Viterbi):** exact best path for the local terms; run per organ independently.

- **Genetic Algorithm:** chromosome = candidate index plus offset per slice (and per organ).
- **Particle Swarm Optimization:** discrete index plus continuous offset per slice.
- **Simulated annealing** and **beam search:** strong non-nature-inspired optimizers; reviewers increasingly ask for these.
- **Random search:** sanity check.
- **ACO ablations:** Ant System vs MMAS; with/without ACO_R refinement; with/without DP seeding; one colony per organ separately vs competing colonies; with/without pheromone memory.

7. Selected AI / Deep Learning Models

MedSAM does the final segmentation and stays frozen; a small 2D nnU-Net suggests where the organ is, and a tiny shape autoencoder judges whether the 3D result looks anatomically normal.

Model	Job in the pipeline	Trained by us?	Why this model
MedSAM (ViT-B)	Turns each candidate box into a precise 2D mask and gives a confidence score	No, frozen	Strong medical foundation model, box-driven, widely cited
nnU-Net 2D	Proposer: gives a rough probability map per slice, from which candidate boxes are made	Yes, once, on one source hospital	The standard, self-configuring U-Net for medical segmentation
Shape autoencoder	Scores how normal the stacked 3D mask looks (low reconstruction error = plausible shape)	Yes, on source-hospital masks only	Tiny and fast; gives a label-free anatomy check
LiteMedSAM (optional)	Faster replacement for MedSAM on a laptop	No	Built for laptop inference

How MedSAM works, simply: it has a large image encoder that turns a slice into an embedding (slow, run once per slice) and a small decoder that takes the embedding plus a box and returns a mask (fast, milliseconds). Because the encoder runs only once, trying many boxes is cheap.

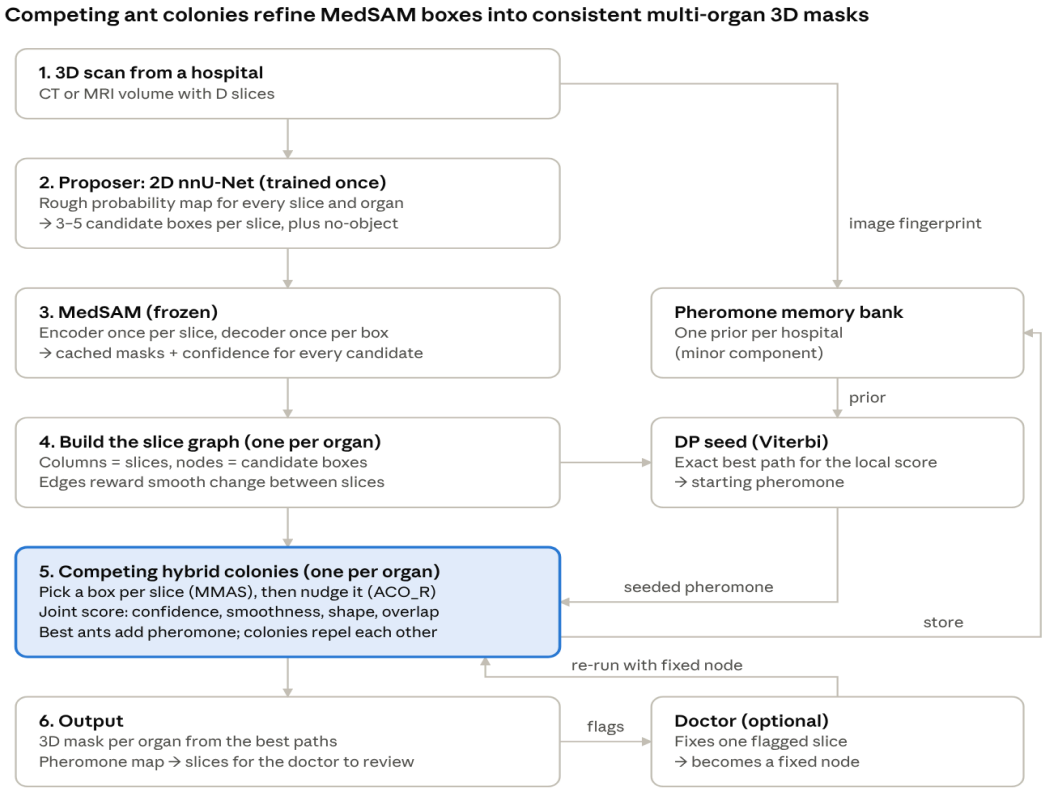
MedSAM2 in plug-in mode. MedSAM2 segments a whole volume from boxes on a few keyframes, and its result depends heavily on which slices are prompted. In plug-in mode the colony's graph changes: nodes are *(slice, box)* pairs for possible keyframes, an ant picks a small set of keyframes (for example 1–3) with their boxes, MedSAM2 propagates, and the same label-free score judges the 3D result. This shows the method is a layer that improves any promptable model, not a competitor to one model. Because each evaluation reruns MedSAM2's propagation, plug-in mode uses fewer ants and rounds and is run on the GPU (Colab/Kaggle).

Models we compare against (not part of our method)

- **MedSAM2:** box on the middle slice, then memory-based propagation through the volume. Compared against MedSAM2 with colony-chosen keyframes.
- **MedSAM with nnU-Net boxes, slice by slice:** the simplest automatic pipeline.
- **nnU-Net 3D:** fully supervised specialist, as a reference.

- **Oracle MedSAM:** boxes taken from the true masks, as an upper bound for box-driven methods.

8. Proposed Architecture / Pipeline



PheroSAM-3D pipeline · 6 steps + pheromone memory

Step 1 — Input and fingerprint. A 3D scan arrives from a hospital. We compute a short fingerprint of its images (intensity histogram statistics plus mean/std of nnU-Net encoder features). The fingerprint is used to look up the pheromone memory.

Step 2 — Candidate boxes. The 2D nnU-Net gives a probability map per slice. From each map we build K = 3–5 candidate boxes using different thresholds (e.g. 0.3, 0.5, 0.7) and margins, plus a "no object" option for slices where the organ is absent. Each candidate keeps a *type* label (which threshold and margin made it), used later by the memory.

Step 3 — MedSAM, once. MedSAM's encoder runs once per slice. Its decoder runs once per candidate box, giving a mask and a predicted-IoU confidence. For a 100-slice scan with 5 candidates that is 100 encoder passes and 500 decoder passes; everything is cached.

Step 4 — Slice graph. Slice d is column d; each candidate box is a node. An edge connects every node in slice $d-1$ to every node in slice d. Its heuristic value κ is the overlap (IoU) between the two masks, because real organs change gradually.

Step 5 — Ant colony search. Pheromone τ sits on edges. An ant at box i in slice $d-1$ chooses box j in slice d with probability:

$$p_{ij}^{(d)} = \frac{\left(\tau_{ij}^{(d)}\right)^{\alpha}\left(\eta_{j}^{(d)}\right)^{\beta}\left(\kappa_{ij}^{(d)}\right)^{\gamma}}{\sum_{l}\left(\tau_{il}^{(d)}\right)^{\alpha}\left(\eta_{l}^{(d)}\right)^{\beta}\left(\kappa_{il}^{(d)}\right)^{\gamma}}$$

η is MedSAM's confidence in box j; α, β, γ set how much each factor matters. A finished path π is scored with the label-free fitness:

$$F(\pi) = w_1\, Conf(\pi) + w_2\, Smooth(\pi) + w_3\, Shape(\pi) + w_4\, Agree(\pi) - w_5\, Pen(\pi)$$

Term	Plain meaning	How it is computed
Conf	MedSAM is sure of its masks	Mean predicted IoU over chosen boxes
Smooth	No sudden jumps between slices	Mean IoU of consecutive masks
Shape	The 3D result looks like a real organ (global term)	Low reconstruction error of the shape autoencoder
Agree	Result agrees with the proposer, but not blindly	Soft Dice with the nnU-Net probability map (small weight)
Pen	No broken or impossible results	Penalty for gaps, several disconnected pieces, or extreme volume

After each round, pheromone evaporates and only the best ant deposits, within MMAS limits:

$$\tau_{ij}^{(d)} \leftarrow clip\Big((1-\rho)\,\tau_{ij}^{(d)} + \Delta\tau_{ij}^{(d)},\ \tau_{min},\ \tau_{max}\Big), \quad \Delta\tau_{ij}^{(d)} = F\left(\pi^{best}\right) if\ edge\ (i,j) \in \pi^{best}$$

The Shape and Pen terms depend on the whole volume, not only on neighbouring slices. That is why exact dynamic programming cannot optimize F directly, while ants can.

Step 5a — DP seed. Before the first round, Viterbi finds the exact best path for the local score (Conf + Smooth). Its edges start at τ_max; all other edges start at τ_seed < τ_max, so ants begin near a good solution but can still leave it.

Step 5b — Continuous refinement. After choosing box j on slice d, the ant adds an offset sampled from the slice's ACO_R archive A_d of the best k offsets found so far:

$$\delta \sim N\left(\delta_r,\ \sigma_r^2\right), \quad r\ chosen\ from\ A_d\ with\ probability\ \propto \omega_r, \quad \sigma_r = \xi \cdot \frac{1}{k-1}\sum_{e \in A_d}\left|\delta_e - \delta_r\right|$$

ω_r is larger for better-ranked offsets, and ξ controls how widely the ants explore. In plain words: good nudges are copied with a little noise, and the noise shrinks as the colony agrees.

Step 5c — Competing colonies. With several organs, colonies take turns. Colony o scores its ants with the joint objective from Section 3, holding the other colonies' current best masks fixed. Repulsive pheromone lowers the heuristic of boxes that overlap another colony's best mask:

$$\eta_j^{(d,o)} \leftarrow \eta_j^{(d,o)} \cdot \left(1 - IoU\left(M_j^{(d)},\ \bigcup_{o' \neq o} M_{best}^{(d,o')}\right)\right)^{\zeta}$$

Pheromone memory (inside step 5). Raw pheromone cannot be copied between patients because they have different numbers of slices. So we store a compact *prior*: for 10 bins of relative slice position (0 = first slice with the organ, 1 = last) we keep how much pheromone went to each candidate type. For a new patient from a known hospital, the colony starts from this prior instead of a flat start; for an unknown hospital it starts flat and a new memory entry is created. After each patient the matching entry is updated with a moving average.

$$\tau^{init} = (1 - \lambda)\,\tau_{max} + \lambda\,\bar{\tau}^{memory}_{k^*}, \quad k^* = \arg\,arg\,\|f - f_k\|$$

Step 6 — Output and uncertainty. The best path gives the final 3D mask. For each slice we measure how spread the pheromone is across its candidates (entropy). High entropy means the colony disagreed, so that slice is flagged for review.

Step 7 — Doctor correction that spreads. The doctor sees the top-k flagged slices (k = 3 by default) and fixes the worst one by drawing a box. That slice becomes a **fixed node**: its edges are clamped at τ_max, every other choice on that slice is removed, and the colonies re-run for a few rounds from their current pheromone. Because the score is global, a single fix can pull neighbouring slices and even neighbouring organs into place. We repeat for 1, 2, 3, 5 and 10 corrections and plot Dice against the number of corrections.

Step 8 — Plug-in mode for MedSAM2. Instead of one box per slice, ants choose a few keyframes and their boxes; MedSAM2 propagates from them and the same score judges the volume (Section 7).

Pseudocode

```
for each volume V from hospital h:
    f <- fingerprint(V)
    for each organ o:
        boxes[o]       <- nnUNet_candidates(V, o)        # K per slice + no-object
        masks[o], eta  <- MedSAM(V, boxes[o])            # encoder once/slice, cached
        kappa[o]       <- IoU of neighbouring-slice masks
        tau[o]         <- DP_seed(Viterbi(local score)) blended with memory(f)
        archive[o]     <- empty ACO_R offset archives, one per slice
    fixed <- {}                                          # doctor corrections
    repeat until doctor stops:
        repeat T rounds:
            for each organ o (in turn):
                for a in 1..m ants:
                    path   <- walk slices, choose box j with p_ij (repulsion applied)
                    path   <- add ACO_R offsets; decode new boxes (cached)
                    score  <- joint fitness(path, best paths of other organs, fixed)
                update tau[o] (MMAS), update archive[o] with best offsets
        masks3D <- best path per organ
        flags   <- slices with highest pheromone entropy
        if doctor corrects slice d of organ o: fixed <- fixed + (o, d, box)
    update_memory(f, tau)
```

Starting parameter values (tuned once on validation data, then frozen)

Parameter	Start value
Ants m (per colony)	20
Rounds T	30, with early stopping

Candidates per slice K	5
α (pheromone weight)	1
β (MedSAM confidence weight)	2
γ (smoothness weight)	1
Evaporation ρ	0.1
DP seed level τ_seed	0.5 × τ_max
ACO_R archive size k	10 per slice
ACO_R spread ξ	0.85
Overlap weight μ / anatomy weight ν	tuned on validation
Repulsion strength ζ	2
Memory mixing λ	0.5
Flagged slices shown to doctor	3

9. Evaluation metrics

Metrics

What we measure	Metric
Accuracy of the 3D mask	3D Dice, HD95 (boundary error), per organ and for tumours separately
Smoothness between slices	Slice Area Consistency Error, number of slice-to-slice jumps
Beating the proposer	Candidate recall (share of slices with at least one good box) and gain over oracle selection
Multi-organ consistency	Overlapping voxels between organs, number of broken anatomical rules
Correction efficiency	Dice after 1, 2, 3, 5, 10 simulated corrections; corrections needed to reach a target Dice
When ACO beats DP	Gap between DP and DP-seeded ACO as the global-term weight increases
Does the label-free score work?	Spearman correlation between F and true Dice
Does the uncertainty help doctors?	Share of truly bad slices caught in the top-k flagged slices
Cost	Seconds per volume, MedSAM decoder calls, RAM/VRAM

Statistics: Wilcoxon signed-rank test per volume, Holm correction, mean ± std over at least 3 random seeds.

10. References

1. Ma et al., Segment Anything in Medical Images (MedSAM), Nature Communications, 2024.
2. Ma et al., MedSAM2: Segment Anything in 3D Medical Images and Videos, 2025.
3. Wu et al., Medical SAM Adapter, Medical Image Analysis, 2025.
4. RadSAM: Segmenting 3D radiological images with a 2D promptable model, 2025.
5. SAMI3D-DW: Interactive Segmentation of Any 3D Medical Images, 2026.
6. Wang et al., SAM-Med3D, 2023.
7. MemSAM-2.5D for 3D liver tumor segmentation, 2026.

8. Automatic Segmentation of 3D CT scans with SAM2 using a zero-shot approach, 2026.
9. Wu et al., SAM-driven prompt-guided delineation of rectal GTV, Medical Physics, 2026.
10. Li et al., Optimal surface segmentation in volumetric images — a graph-theoretic approach, IEEE TPAMI, 2006.
11. Attack for Defense: Adversarial Agents for Point Prompt Optimization for SAM, 2025 (survey of auto-prompting methods).
12. Enhancing MedSAM with a Lightweight Box Predictor, 2026.
13. Multiple Point MedSAM Prompting, 2026.
14. Shen et al., Temporally-Extended Prompts Optimization for SAM (TEPO), 2023.
15. Enhanced ACO for multi-threshold COVID-19 X-ray segmentation, 2023.
16. EvoU-Net: evolutionary U-Net for medical image segmentation, 2020.
17. PSO-Based Transfer Learning for Image Segmentation and Classification, 2026.
18. Niu et al., Test-Time Model Adaptation with Only Forward Passes (FOA), ICML 2024.
19. Zhang et al., DPCore: Dynamic Prompt Coreset for Continual Test-Time Adaptation, ICML 2025.
20. Dorigo & Stützle, Ant Colony Optimization, MIT Press, 2004; Stützle & Hoos, MAX–MIN Ant System, Future Generation Computer Systems, 2000; Isensee et al., nnU-Net, Nature Methods, 2021 (from memory; verify).
21. Socha & Dorigo, Ant colony optimization for continuous domains (ACO_R), European Journal of Operational Research, 2008.
22. Middendorf, Reischle & Schmeck, Multi colony ant algorithms, Journal of Heuristics, 2002.
23. Ji et al., AMOS: a large-scale abdominal multi-organ benchmark, NeurIPS Datasets and Benchmarks, 2022.
24. Antonelli et al., The Medical Segmentation Decathlon, Nature Communications, 2022.
25. Campello et al., Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation: the M&Ms Challenge, IEEE TMI, 2021.